

A Brief Review of Remote Sensing Basics

Alyssa Willson

2026-03-06

RS is really the study of light (electromagnetic radiation)

- YouTube video: Is Light a Particle or a Wave?
- Light is unlike other stuff we encounter in the world.
- What we can actually measure is the amount of light (watts per square meter) in a given solid angle in a given wavelength range
- This is called “radiance”

Active vs Passive Remote Sensing

- **Active** sensors emit EMR and then measure how long it takes to get back to the sensor (e.g. lidar, radar)
- **Passive** sensors collect EMR that is either coming from the sun and reflected off an object (e.g. multi- and hyperspectral) or collect EMR that is emitted from an object (e.g. thermal, SIF)

Remote Sensing (RS) “Platforms”

- The ground (“near surface RS”)
- Drones/UAVs
- Airplanes
- Satellites
 - Tasked
 - “Full Coverage”

Radiative Transfer

- **EMR behaves differently as it passes through or bounces off different materials**
- EMR behaves differently **depending on the wavelength** in question
- Accounting for these differences is called **Radiative Transfer Modelling**, literally the estimation of EMR transferring from one medium to another.
- Virtually all remote sensing is aimed at trying to understand how EMR interacts with, is emitted by, or bounces off stuff (plants, rocks, the atmosphere, clouds, lava, water, etc).

Radiance and Reflectance

- **We measure radiance but we care about reflectance.**

Types of Remote Sensing

- The **VAST** majority of terrestrial remote sensing uses **passive optical** multispectral spaceborne sensors, like Landsat, MODIS, and Sentinel-2
 - = Optical is not the same as visible. Optical includes the visible, NIR, SWIR, and thermal

- Terrestrial remote sensing focused on structure uses active sensing like lidar (in optical wavelengths, mostly airborne) or radar (radio & microwaves, mostly spaceborne)
- NEON's Airborne Observation Platform (AOP) includes a passive optical hyperspectral

Scale & Resolution

- It's more than just pixel size!
- Spatial extent and grain size (fine vs. coarse)
- Temporal extent and frequency
- Spectral extent and resolution
- Radiometric and technical specs

VSWIR Spectroscopy

- VIS: What we see; plant function
- NIR: Cell structure; plant/soil brightness; plant chemistry
- SWIR: Water content across materials; plant chemistry

VSWIR Spectroscopy—contact spectroscopy

- Also, these spectral signatures have gone through atmospheric correction!

Landsat & Sentinel-2

Coarse Spatial Resolution Satellites (MODIS & VIIRS)

- Multispectral like Landsat & Sentinel-2, but much coarser spatial resolution

Common Methods for Analyzing Multi/Hyperspectral Data

Spectral Indices

- **Normalized Difference Vegetation Index (NDVI)**

$$= \frac{NIR - red}{NIR + red}$$
- Lots of other indices—question is always WHAT bands do they use, where were they developed, and what sensor(s) were used?

Land Cover Classification / Clustering

Trait Mapping with Statistical Approaches

- Collect leaf samples (& GPS locations)
- Collect leaf or image spectra
- Process leaves for traits in lab
- Assemble & clean data!
- Partial least squares regression (PLSR)
- Predict traits of unknown leaves/pixels

VSWIR Spectroscopy Wrap-Up

- Spectral Indices are common ways of simplifying multi and hyperspectral data.
- Classification/Clustering is the most common way of turning spectral (and other) data into a useable map.
- PLSR is the “standard” technique for mapping plant traits with hyperspectral data—we will spend more time on this later in the course
- Many other techniques for processing/thinking about spectral data!

Questions on Spectroscopy?

Light Detection & Ranging (lidar)

- YouTube video: How Does LiDAR Remote Sensing Work? Light Detection and Ranging

Wavelengths and their Properties

Point Cloud Data

- X, Y, Z coordinates
- Intensity
- Sometimes other information
- Either .las or .laz file formats

Processing Point Clouds

- From the Warra SuperSite in Tasmania, Australia
- Don't forget the third dimension!

Waveform Lidar

CHM & DEM

- Canopy Height Model (CHM) & Digital Elevation Model (DEM)

Within-Canopy Metrics

Lidar Wrap-Up

- Lidar data depends a lot on the specifications of the system
- Most vegetation-focused systems use 1064 nm light
- Main goal is usually to map canopy height (CHM) and elevation (DEM)
- There are LOTS of metrics for calculating within canopy shape/density

Questions on lidar?

NEON AOP Data Applications

Sources of Uncertainty

- Physics and engineering
- Phenology and physiological change over time
 - ORNL PhenoCam July 2020
- Spectral Unmixing in coarse-ish data
- Under-documented parts of the world

Acknowledgments

- Feedback on this lecture was provided by Keely Roth (University of Utah & Planet Labs), Fred Huemmrich (UMBC and NASA), and Adriana Uscanga (UMN)
- SPEC School is supported by NSF DEB Award #2044818